```
[swaraj@Swaraj weather_log-mapreduce]$ cd input
[swaraj@Swaraj input]$ ls
sample_weather.txt
[swaraj@Swaraj input]$ cd ..
[swaraj@Swaraj weather_log-mapreduce]$ cat mapper.py
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    parts = line.strip().split()
    if len(parts) == 5:
        temp = parts[2]
        dew = parts[3]
        wind = parts[4]

        print(f"TEMP\t{temp}")
        print(f"DEW\t{dew}")
        print(f"WIND\t{wind}")
[swaraj@Swaraj weather_log-mapreduce]$ cat reducer.py
#!/usr/bin/env python3
import sys

data = {}

for line in sys.stdin:
    key, value = line.strip().split("\t")
    value = float(value)

    if key not in data:
        data[key] = [value, 1]
    else:
        data[key][0] += value
        data[key][1] += 1

for key in data:
    total, count = data[key]
    avg = total / count
    print(f"{key}_avg\t{avg}")
[swaraj@Swaraj weather_log-mapreduce]$ hadoop jar $HADOOP_HOM
.py
2026-04-18 23:43:09,818 INFO impl.MetricsConfig: Loaded prope
2026-04-18 23:43:09,888 INFO impl.MetricsSystemImpl: Schedule
2026-04-18 23:43:09,888 INFO impl.MetricsSystemImpl: JobTrack
2026-04-18 23:43:09,898 WARN impl.MetricsSystemImpl: JobTrack
2026-04-18 23:43:10,006 INFO mapred.FileInputFormat: Total in
2026-04-18 23:43:10,017 INFO mapreduce.JobSubmitter: number o
2026-04-18 23:43:10,142 INFO mapreduce.JobSubmitter: Submitti
2026-04-18 23:43:10,142 INFO mapreduce.JobSubmitter: Executin
```

```
2026-04-18 23:43:10,637 INFO mapred.LocalJobRunner: Finishing task: attempt_loc
2026-04-18 23:43:10,637 INFO mapred.LocalJobRunner: reduce task executor comple
2026-04-18 23:43:11,281 INFO mapreduce.Job: Job job_local29493010_0001 running
2026-04-18 23:43:11,283 INFO mapreduce.Job:  map 100% reduce 100%
2026-04-18 23:43:11,285 INFO mapreduce.Job: Job job_local29493010_0001 complete
2026-04-18 23:43:11,295 INFO mapreduce.Job: Counters: 30
        File System Counters
                FILE: Number of bytes read=287872
                FILE: Number of bytes written=1564570
                FILE: Number of read operations=0
                FILE: Number of large read operations=0
                FILE: Number of write operations=0
        Map-Reduce Framework
                Map input records=45
                Map output records=135
                Map output bytes=1035
                Map output materialized bytes=1311
                Input split bytes=116
                Combine input records=0
                Combine output records=0
                Reduce input groups=3
                Reduce shuffle bytes=1311
                Reduce input records=135
                Reduce output records=3
                Spilled Records=270
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=9
                Total committed heap usage (bytes)=515899392
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=1170
        File Output Format Counters
                Bytes Written=95
2026-04-18 23:43:11,295 INFO streaming.StreamJob: Output directory: output
[swaraj@Swaraj weather_log-mapreduce]$ cat output/part-00000
DEW_avg 22.666666666666668
TEMP_avg        32.666666666666664
WIND_avg        18.466666666666665
[swaraj@Swaraj weather_log-mapreduce]$
```